

AI Tools Virtual Lab

Complete Syllabus Reference

5 Weeks • 5 Modules • 20 Experiments

Image Generation • Writing • Coding • Web Design • Research & Productivity

Aims • ELI5 Theory • Flows • Comparison Tables • Image Prompts

Designed so anyone — from school student to professional — can understand it.

Table of Contents

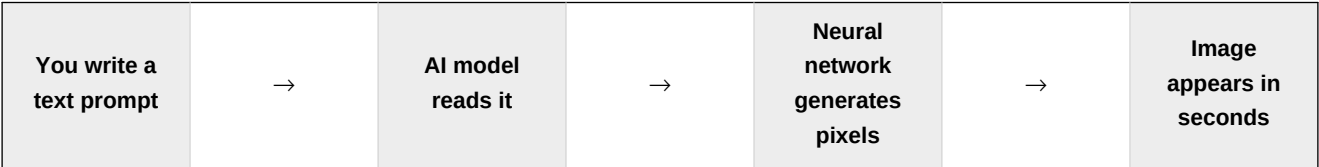
W1	Image Generation Lab	
	Exp 1	Gemini Imagen Photorealism
	Exp 2	DALL-E Concept Art
	Exp 3	Leonardo AI Character Design
	Exp 4	Midjourney Artistic Composition
W2	Writing Lab	
	Exp 1	ChatGPT Persuasive Essay
	Exp 2	Claude Paper Summary
	Exp 3	Gemini Blog Post
	Exp 4	Perplexity Research Query
W3	Coding Lab	
	Exp 1	Claude Code Mini-Project
	Exp 2	Cursor AI Refactoring
	Exp 3	GitHub Copilot Productivity
	Exp 4	Replit Agent Prototyping
W4	Web Design Lab	
	Exp 1	Lovable Landing Page
	Exp 2	Bolt.new Web App
	Exp 3	v0 Component Library
	Exp 4	Figma AI Layouts
W5	Research & Productivity Lab	
	Exp 1	NotebookLM Document Chat
	Exp 2	Gamma Presentation Deck
	Exp 3	Napkin AI Diagrams
	Exp 4	Elicit Literature Review

Week 1 ■ Image Generation Lab

Module 1: Visual AI Basics | Lab 1: Concept to Canvas

■ MODULE OVERVIEW

Text-to-image AI models take plain English descriptions (called prompts) and generate photorealistic or artistic images in seconds. Each tool has a different strength — photorealism, concept art, characters, or painterly styles. In this lab we explore four of the best.



Experiment 1: Gemini Imagen Photorealism

■ AIM

To use Google's Imagen model via the Gemini platform to generate highly photorealistic images from detailed text prompts, exploring how prompt specificity, lighting descriptors, and subject detail directly influence output quality and realism.

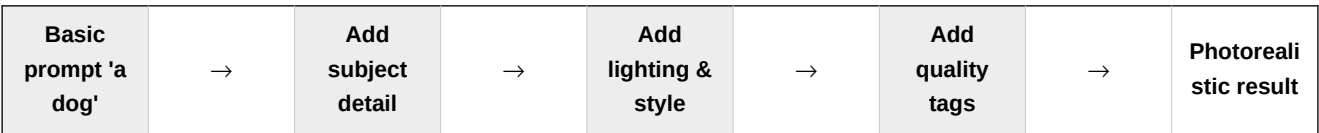
■ THEORY

What Makes Imagen Special?

Imagen is Google's text-to-image model built for photorealism — outputs that look like actual photographs rather than illustrations. It was trained on billions of image-text pairs, learning to connect words like 'golden hour lighting' or 'shallow depth of field' to real visual qualities.

Prompt Element	What It Controls	Example
Subject	What is in the image	'A golden retriever sitting on grass'
Lighting	Mood and atmosphere	'soft morning light', 'studio lighting'
Camera style	Depth and framing	'shallow depth of field', '35mm lens'
Quality tag	Forces high detail	'ultra-realistic', '8K', 'DSLR photo'

Prompt Engineering for Images



■ IMAGE PROMPT

Soft pastel flat infographic: left side shows a simple text box with 'a dog' producing a blurry sketch; right side shows a detailed prompt with lighting and quality tags producing a sharp photo; arrow flow between them labeled 'prompt engineering'; checklist of prompt elements on the side; cream background; minimal text; friendly educational style.

Experiment 2: DALL-E Concept Art

■ AIM

To leverage OpenAI's DALL-E model to generate stylised concept art by experimenting with artistic style descriptors, medium references, and compositional framing instructions, understanding how abstract creative direction translates into visual output.

■ THEORY

DALL-E and Artistic Styles

DALL-E excels at concept art and stylised illustration. You can tell it to paint like a specific era, medium, or artistic movement and it will blend those aesthetics into a single image.

Style Descriptor	What You Get
'watercolour illustration'	Soft, blended painterly look
'oil painting, baroque style'	Rich, dramatic, classical art
'flat vector art'	Clean minimal graphic design
'cyberpunk concept art'	Neon, futuristic, gritty urban scenes
'Studio Ghibli style'	Warm, hand-drawn anime aesthetic

The Concept Art Workflow

Define the concept / idea	→	Choose an artistic style	→	Add mood & composition	→	Iterate with variations	→	Final concept art
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■ IMAGE PROMPT

Soft pastel infographic: four small image frame boxes each showing a different art style label (watercolour, oil painting, vector, cyberpunk); arrows pointing from one text prompt to all four frames; label 'same prompt, different style'; cream background; minimal text; flat educational style.

Experiment 3: Leonardo AI Character Design

■ AIM

To use Leonardo AI's fine-tuned models to design consistent, detailed characters by crafting structured character description prompts covering appearance, costume, emotion, and pose — and to explore how model selection affects stylistic consistency across multiple character renders.

■ THEORY

Why Leonardo for Characters?

Leonardo AI offers specialised fine-tuned models — versions trained specifically on character art, game assets, or anime. This means you get much more consistent results for the same character across multiple images compared to general-purpose models.

Character Prompt Layer	What to Describe
Appearance	Hair colour, eye colour, skin tone, age, body type
Costume	Clothing style, colours, materials, accessories
Emotion	'Determined expression', 'joyful smile', 'fierce gaze'
Pose	'Standing heroically', 'crouching', 'arms crossed'
Background	'Plain white', 'fantasy forest', 'studio backdrop'

Consistency Across Renders



■ IMAGE PROMPT

Soft pastel infographic: a character design sheet showing the same character in three poses (standing, action, close-up portrait) with labels pointing to appearance, costume, emotion, pose; flat illustration style; cream background; minimal text; educational poster layout.

Experiment 4: Midjourney Artistic Composition

■ AIM

To use Midjourney's advanced compositional controls — including aspect ratio flags, stylisation parameters, and negative prompts — to craft intentional artistic compositions that demonstrate mastery of visual balance, colour palette direction, and aesthetic coherence.

■ THEORY

Midjourney's Unique Parameters

Midjourney uses special command flags alongside your prompt to control the technical and artistic properties of the output.

Parameter	What It Does	Example
--ar	Sets aspect ratio	--ar 16:9 (widescreen), --ar 1:1 (square)
--stylize	Controls artistic intensity	--stylize 100 (subtle) to 1000 (very painterly)
--no	Negative prompt — excludes	--no text, --no people, --no blur
--v	Model version	--v 6 (latest, most detailed)
--chaos	Variation randomness	--chaos 0 (consistent) to 100 (wild)

Building a Composed Prompt



■ IMAGE PROMPT

Soft pastel flat infographic: a prompt formula diagram showing subject plus mood plus style plus parameters all combining with plus signs into one final image frame; parameter cheat sheet in a side box (ar, stylize, no, chaos); cream background; minimal text; clean educational layout.

Week 2 🖋️ Writing Lab

Module 2: Generative Text | Lab 2: The AI Co-Writer

MODULE OVERVIEW

AI writing tools can draft essays, summarise papers, write blog posts, and search the web for research — in seconds. Each tool has a personality: ChatGPT is creative, Claude is precise, Gemini is structured, Perplexity cites sources.

Tool	Best At	Style
ChatGPT	Persuasive and creative writing	Engaging, conversational
Claude	Summarisation and analysis	Precise, careful, structured
Gemini	Long-form drafting and research	Clear, Google-powered
Perplexity	Research with live web citations	Factual, source-linked

Experiment 1: ChatGPT Persuasive Essay

AIM

To use ChatGPT to draft, refine, and strengthen a persuasive essay by designing targeted prompts that specify audience, argument stance, tone, and structural requirements — demonstrating how iterative prompt refinement improves argument quality and rhetorical impact.

THEORY

What Makes Writing Persuasive?

Rhetorical Element	What It Means	Prompt Instruction
Ethos	Build credibility	'Use expert evidence and statistics'
Pathos	Appeal to emotion	'Include a relatable human story'
Logos	Use logic and reason	'Structure argument with clear evidence'
Kairos	Right time / urgency	'Highlight why this matters right now'

The Iterative Writing Flow

Give topic + stance	→	First draft generated	→	Ask: 'strengthen the argument'	→	Ask: 'add statistics'	→	Final polished essay
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IMAGE PROMPT

Soft pastel infographic: a document icon being refined through three iterations with checkmarks; side panel showing the three rhetorical elements (ethos pathos logos) as labelled building blocks; arrow loop showing iterative prompting; cream background; minimal text; friendly flat style.

Experiment 2: Claude Paper Summary

AIM

To use Claude to produce accurate, structured summaries of academic papers by constructing prompts that specify summary depth, target audience, and output format — evaluating Claude's ability to extract key findings, methodology, and implications without distortion or hallucination.

■ THEORY

Why Claude for Academic Summarisation?

Claude is trained to be precise and faithful to source material. It is less likely to add information that wasn't in the paper, making it safer for academic use than more creative models.

Summary Type	Prompt Instruction	Output
Executive	'Summarise in 3 bullet points for a manager'	Quick top-level overview
Technical	'Summarise the methodology and results for a researcher'	Deep, precise, jargon-intact
Plain English	'Explain this paper to a 16-year-old student'	Simple, analogies, no jargon

The Summarisation Flow



■ IMAGE PROMPT

Soft pastel: a dense academic paper on the left being fed into a robot (Claude) who outputs three clean summary cards labeled executive, technical, plain English; magnifying glass icon showing faithful extraction not invention; cream background; minimal text.

Experiment 3: Gemini Blog Post

■ AIM

To use Google Gemini to generate a well-structured, SEO-aware blog post by providing prompts that define topic, target keyword, reader persona, tone, and desired word count — and to iteratively refine headings, introductions, and calls-to-action for maximum engagement.

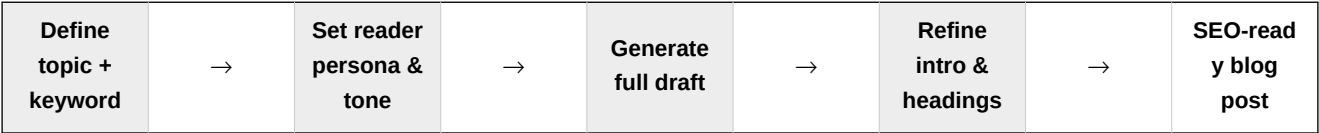
■ THEORY

Anatomy of a Great Blog Post

Section	Purpose	Prompt Instruction
Hook / Intro	Grab attention immediately	'Start with a surprising statistic or question'
H2 Headings	Structure for scanning readers	'Use 4 H2 subheadings with keyword'
Body Paragraphs	Deliver the value	'Each section: explain, example, takeaway'
Call to Action	Tell reader what to do next	'End with a clear CTA for the reader'

SEO Awareness in Prompts

SEO = Search Engine Optimisation. A well-prompted blog post naturally includes the target keyword in the title, first paragraph, and subheadings — making it more likely to appear in Google search results.



■ IMAGE PROMPT

Soft pastel infographic: a blog post wireframe with labeled sections (hook, H2 headings, body, CTA); keyword highlighted with a magnifying glass labeled SEO; robot writing at a desk; iteration arrows showing refinement; cream background; minimal text; flat friendly style.

Experiment 4: Perplexity Research Query

■ AIM

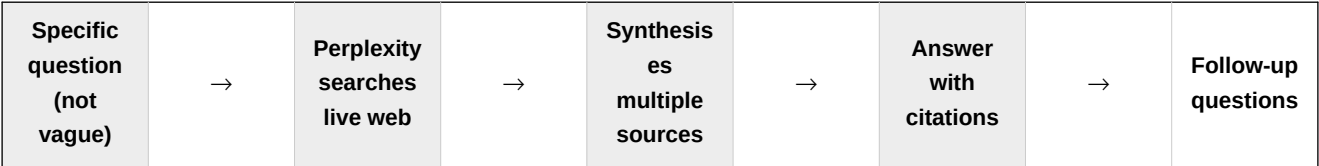
To use Perplexity AI as an AI-powered research engine by formulating precise research queries that leverage its live web search capability, evaluating source citation quality, assessing answer accuracy across factual and contested topics, and comparing its research output to traditional search engine results.

■ THEORY

Perplexity vs Google — What's Different?

	Google Search	Perplexity AI
What you get	List of links to click	Synthesised answer with citations
Time to insight	You read multiple articles	Instant summary from multiple sources
Citations	Implicit (you check yourself)	Explicit inline source links
Best for	Finding specific pages	Research questions needing synthesis

Writing a Good Research Query



■ IMAGE PROMPT

Soft pastel infographic: left side shows Google returning a list of blue links; right side shows Perplexity returning one clear answer with numbered citation bubbles; label 'links vs synthesis'; a magnifying glass over the web at the top; cream background; minimal text; flat educational style.

Week 3 ■ Coding Lab

Module 3: AI in Software Development | Lab 3: The AI Pair Programmer

■ MODULE OVERVIEW

AI coding assistants can write code from scratch, fix bugs, refactor messy code, and prototype full apps in minutes. They act as a tireless pair programmer who knows every programming language and framework.

Tool	Best At	How It Works
Claude Code	Full project reasoning & architecture	Chat-based, file-aware coding agent
Cursor AI	Refactoring and editing existing code	VS Code fork with AI deeply embedded
GitHub Copilot	Line-by-line autocomplete as you type	Inline suggestion inside your editor
Replit Agent	Full-stack prototyping in the browser	Build and run apps without any setup

Experiment 1: Claude Code Mini-Project

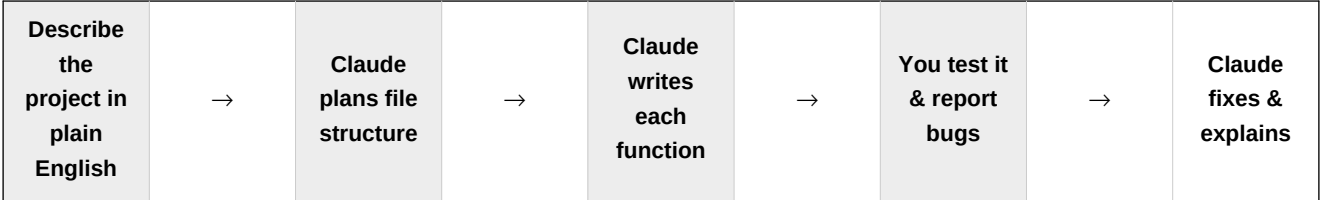
■ AIM

To use Claude as an AI software architect by providing high-level project briefs and iteratively decomposing them into working code — covering file structure planning, function-by-function implementation, and debugging cycles — demonstrating end-to-end AI-assisted software development from specification to running output.

■ THEORY

Claude as a Software Architect

Unlike autocomplete tools, Claude understands the full context of your project. You describe what you want to build in plain English and Claude plans the structure, writes each part, and explains every decision.



Effective Project Prompting

Prompt Element	Example
Goal	'Build a Python script that reads a CSV and outputs a summary'
Constraints	'Use only standard libraries, no pip installs'
Output form	'Show me the file structure first, then write each file'
Debug style	'Here is the error message — explain what went wrong'

■ IMAGE PROMPT

Soft pastel: a human at a desk with a speech bubble describing a project; Claude robot outputting a file tree diagram with arrows to individual code files; then a bug icon with a wrench showing the fix cycle; cream background; minimal text; flat style.

Experiment 2: Cursor AI Refactoring

■ AIM

To use Cursor AI's editor-integrated AI to refactor existing codebases by applying targeted AI instructions for improving code readability, reducing duplication, enforcing naming conventions, and splitting monolithic functions — measuring before-and-after code quality improvements.

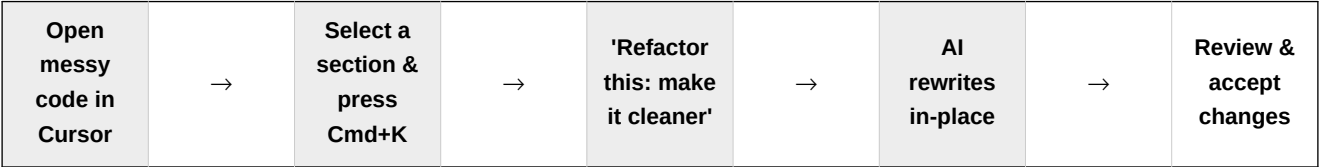
■ THEORY

What Is Refactoring?

Refactoring means improving the structure of existing code without changing what it does. Think of it like reorganising a messy room — everything still works, but it's now clean and easy to navigate.

Refactoring Task	Before	After
Rename variables	'x', 'temp', 'data2'	Descriptive names: 'user_age', 'invoice_total'
Split long function	One 200-line function	10 focused 20-line functions
Remove duplication	Same logic copy-pasted 5 times	One reusable function called 5 times
Add comments	No explanations in code	Every function has a clear docstring

Cursor AI Refactoring Flow



■ IMAGE PROMPT

Soft pastel: left side shows tangled spaghetti code lines in a box labeled 'before'; right side shows neat organised code blocks labeled 'after'; a magic wand between them labeled 'Cursor AI refactor'; checklist of refactoring tasks on the side; cream background; minimal text.

Experiment 3: GitHub Copilot Productivity

■ AIM

To measure the productivity impact of GitHub Copilot's inline code completion by completing a set of timed coding tasks with and without Copilot enabled — documenting keystrokes saved, time reduction, and accuracy of suggestions across different programming languages and task types.

■ THEORY

How GitHub Copilot Works

Copilot watches what you type in real time. As you write a function name or a comment describing what you need, it predicts the entire code block and shows it in grey text. Press Tab to accept. It's like autocomplete on steroids.

Trigger Type	What You Type	What Copilot Generates
Comment trigger	# sort this list in descending order	Full sort function with return
Function name	def calculate_tax(	Parameters, logic, docstring
Pattern repeat	First item in a list pattern	All remaining items automatically
Test generation	# write unit tests for this function	Full pytest test file

■ IMAGE PROMPT

Soft pastel: a code editor showing a comment in dark text and ghost grey suggestion appearing below; Tab key highlighted with arrow labeled 'accept'; side bar showing productivity stats (time saved, keystrokes reduced); cream background; minimal text; flat style.

Experiment 4: Replit Agent Prototyping

■ AIM

To use Replit Agent to build, run, and iterate on a full-stack web application prototype entirely in the browser — from natural language specification through to a live deployed URL — without any local environment setup, demonstrating AI-powered zero-configuration rapid prototyping.

■ THEORY

Why Replit Agent Is Different

Most coding tools just write code — you still need to install software, set up a server, and figure out deployment yourself. Replit Agent handles everything: it writes the code, runs it, fixes errors, and gives you a live URL — all in one place.

Step	Traditional Dev	Replit Agent
Environment setup	Install Node, Python, DB — hours	Zero setup — runs in browser
Write code	You write everything manually	Agent writes from your description
Fix errors	You debug line by line	Agent detects and fixes automatically
Deploy to web	Configure servers, DNS, hosting	One click — instant live URL

Describe your app in English	→	Agent writes all the code	→	Runs it automatically	→	Fixes any errors itself	→	Live URL in minutes
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■ IMAGE PROMPT

Soft pastel: a person typing a plain English description into a chat box; Replit robot outputting code files, a terminal window, and a phone showing a live website; timeline below showing traditional dev (many steps, many hours) vs Replit agent (few steps, minutes); cream background; minimal text; flat educational style.

Week 4 ■ Web Design Lab

Module 4: Generative UI | Lab 4: Zero to Website

■ MODULE OVERVIEW

AI web design tools let you go from an idea to a live, styled website or UI component using plain English — no design degree or coding knowledge required. Each tool targets a different part of the process.

Tool	Best For	Output
Lovable	Full landing pages	Complete hosted website from a brief
Bolt.new	Full-stack web apps	Working app with backend and database
v0	React UI components	Copy-pasteable component code
Figma AI	Design mockups & layouts	Design file with AI-generated layouts

Experiment 1: Lovable Landing Page

■ AIM

To use Lovable to generate a complete, styled landing page from a single product brief prompt — iterating on layout, copy, colour scheme, and section structure through natural language instructions — and publish it to a live URL without writing any code.

■ THEORY

Anatomy of a Landing Page

Section	Purpose	What to Tell Lovable
Hero	First impression — hook the visitor	'Bold headline + subheading + CTA button'
Features	Show what it does	'3 feature cards with icons'
Social Proof	Build trust	'Testimonials section with 3 quotes'
Pricing	Convert visitors	'3-tier pricing table'
Footer	Links and legal	'Links, social icons, copyright'

Write product brief in chat	→	Lovable generates full page	→	Ask: 'change colour to blue'	→	Ask: 'add testimonials'	→	Publish to live URL
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■ IMAGE PROMPT

Soft pastel: a chat box with a product description on the left; on the right a rendered landing page wireframe with labeled sections (hero, features, pricing, footer); iteration arrows between chat and page showing refinement; cream background; minimal text.

Experiment 2: Bolt.new Web App

■ AIM

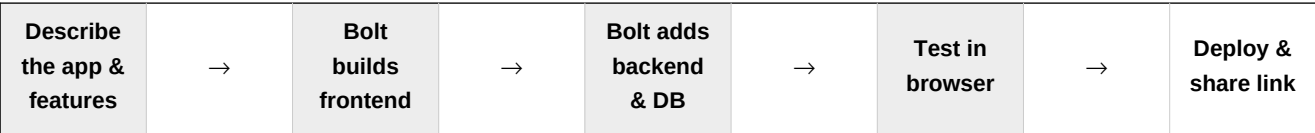
To use Bolt.new to build a complete full-stack web application from a natural language specification — including frontend UI, backend logic, and data persistence — deployed live in the browser, demonstrating

AI-driven full-stack development without any local environment configuration.

■ THEORY

Full-Stack vs Landing Page

	Landing Page (Lovable)	Full-Stack App (Bolt.new)
Stores data?	No	Yes — database included
User accounts?	No	Yes — auth supported
Dynamic?	Static content only	Changes based on user actions
Examples	Marketing site, portfolio	Todo app, booking system, dashboard



■ IMAGE PROMPT

Soft pastel: three-layer stack diagram showing frontend (UI), backend (logic), database (data) all being assembled by a robot from one text prompt; browser window showing the live app; comparison box: landing page (flat) vs web app (layered); cream background; minimal text.

Experiment 3: v0 Component Library

■ AIM

To use Vercel's v0 to generate production-ready React UI components from natural language descriptions — including forms, navigation bars, cards, and modals — and evaluate the generated Tailwind CSS and shadcn/ui code for direct integration into existing React projects.

■ THEORY

What Is a UI Component?

A UI component is a reusable building block of a website — like a button, a login form, or a navigation bar. v0 generates these as clean React code you can copy and paste directly into your project.

Component Type	What You Describe	What v0 Generates
Form	'Sign-up form with name, email, password, submit button'	Full React form with validation
Nav bar	'Dark navbar with logo left, links centre, CTA right'	Responsive navbar component
Card	'Product card with image, title, price, add-to-cart button'	Styled card component
Modal	'Confirmation dialog with cancel and confirm buttons'	Accessible modal with overlay

■ IMAGE PROMPT

Soft pastel: a row of four UI component boxes (form, navbar, card, modal) each labeled; above them a single chat prompt arrow pointing to all four; below each component a code snippet icon showing React output; cream background; minimal text; flat educational layout.

Experiment 4: Figma AI Layouts

■ AIM

To use Figma's AI-powered design features to auto-generate layout variations, apply design system constraints, and produce responsive grid structures from content briefs — evaluating AI-generated designs against established UX heuristics for usability, visual hierarchy, and accessibility.

■ THEORY

What Figma AI Can Do

Figma AI Feature	What It Does
Auto Layout AI	Generates responsive frames that adapt to content size
Design Suggestions	Proposes layout variations based on your content
Component Matching	Finds the right component from your design system automatically
Copy Generation	Fills placeholder text with contextually relevant content

Evaluating AI Designs — UX Heuristics

Heuristic	What to Check
Visual Hierarchy	Is the most important element largest / most prominent?
Whitespace	Is there breathing room or is it too cluttered?
Consistency	Do fonts, colours and spacing match the design system?
Accessibility	Is contrast ratio above 4.5:1? Are tap targets large enough?

■ IMAGE PROMPT

Soft pastel: a Figma canvas frame with AI-generated layout grid; four labelled arrows pointing to design elements labeled hierarchy, whitespace, consistency, accessibility; small checklist box on the side with tick marks; cream background; minimal text; flat style.

Week 5 ■ Research & Productivity Lab

Module 5: AI Workflows | Lab 5: The Augmented Knowledge Worker

■ MODULE OVERVIEW

AI productivity tools augment knowledge work — turning hours of reading, slide-making, diagram-drawing, and literature searching into minutes. These tools don't replace thinking — they accelerate it.

Tool	Superpower	Time Saved
NotebookLM	Chat with your own documents	Hours of re-reading
Gamma	Instant presentation decks	Hours of slide design
Napkin AI	Text to diagrams automatically	Hours of diagramming
Elicit	AI literature search with paper analysis	Days of manual review

Experiment 1: NotebookLM Document Chat

■ AIM

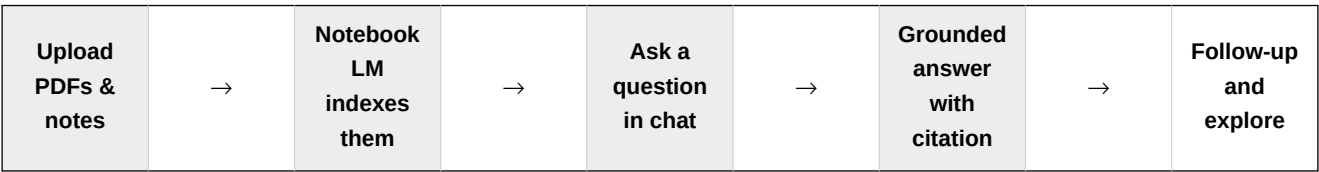
To use Google NotebookLM to ingest multiple source documents and conduct an interactive question-answering session that extracts, cross-references, and synthesises information across sources — evaluating the accuracy of grounded citations and the tool's ability to surface non-obvious connections between documents.

■ THEORY

NotebookLM — Your Personal Research Assistant

NotebookLM works like a very smart friend who has read all your documents. You upload PDFs, notes, or links. Then you ask questions and it answers — citing the exact source and page so you can verify everything.

Feature	What It Does
Grounded answers	Every answer cites the exact document and passage it came from
Cross-referencing	Finds connections between multiple documents automatically
No hallucination	Will not answer from outside your uploaded sources
Audio overview	Generates a podcast-style summary of your documents



■ IMAGE PROMPT

Soft pastel: multiple PDF document icons being fed into a notebook robot; user asking a question in a chat bubble; robot responding with an answer and a citation tag linking back to a specific document; cross-reference arrows between two documents; cream background; minimal text; flat educational style.

Experiment 2: Gamma Presentation Deck

■ AIM

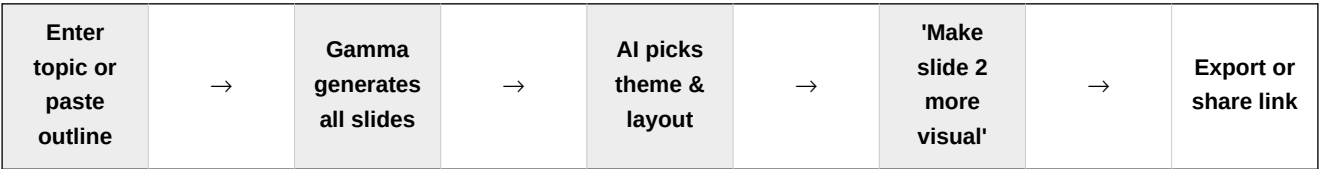
To use Gamma to transform a structured content outline into a visually polished presentation deck in under five minutes — evaluating the AI's slide structuring logic, visual theme selection, and content distribution — and iterating on specific slides through natural language editing instructions.

■ THEORY

What Gamma Does Differently

Traditional slide tools (PowerPoint, Google Slides) give you a blank canvas. Gamma gives you an AI that designs, writes, and layouts each slide for you — you just provide the topic or outline.

	Traditional (PowerPoint)	AI-Powered (Gamma)
Starting point	Blank slide	AI-generated full deck from outline
Design time	Hours choosing fonts and layouts	Seconds — AI picks everything
Content	You write every word	AI drafts, you refine
Editing	Manual drag and drop	Natural language: 'make slide 3 shorter'



■ IMAGE PROMPT

Soft pastel: a plain text outline on the left transforming via a magic wand into a row of styled presentation slide thumbnails on the right; chat bubble showing 'make it more colourful'; timer icon showing 5 minutes; cream background; minimal text; flat friendly style.

Experiment 3: Napkin AI Diagrams

■ AIM

To use Napkin AI to automatically convert written explanations and structured text into clear visual diagrams — including flowcharts, comparison frameworks, and process maps — evaluating the semantic accuracy of AI-generated visuals and iterating on diagram type and layout through natural language instructions.

■ THEORY

Why Diagrams Matter

Research shows people understand and remember information 6x better when it's presented visually. But drawing diagrams manually in tools like Lucidchart or draw.io takes time. Napkin AI generates them from your text in seconds.

Diagram Type	Best For	Example Text Input
Flowchart	Processes and decisions	'User logs in → check password → success or retry'
Comparison	Side-by-side evaluation	'RAG vs Fine-tuning: cost, speed, accuracy'
Timeline	Sequential events	'Phase 1: Research, Phase 2: Build, Phase 3: Test'
Mind Map	Brainstorming relationships	'AI Tools: Writing, Coding, Design, Research'



■ IMAGE PROMPT

Soft pastel: a paragraph of plain text on the left with a magic wand arrow pointing to four different diagram types on the right (flowchart, comparison table, timeline, mind map); label 'text in, diagram out'; cream background; minimal text; flat educational style.

Experiment 4: Elicit Literature Review

■ AIM

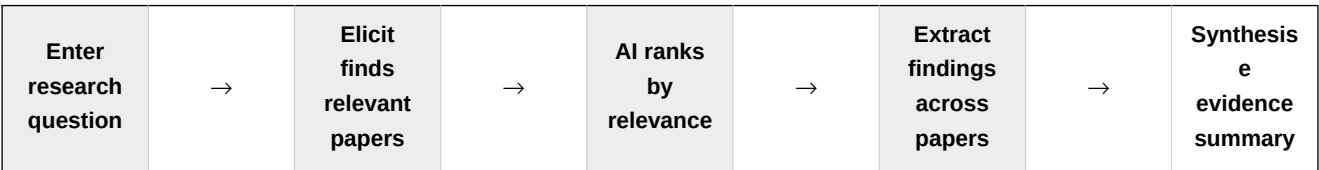
To use Elicit as an AI research assistant for conducting a structured literature review by submitting targeted research questions, evaluating the relevance ranking of returned papers, extracting key findings across multiple studies using Elicit's column-based extraction system, and synthesising a coherent evidence summary from the results.

■ THEORY

What Is a Literature Review?

A literature review means reading many academic papers on a topic to understand what researchers have already discovered. Traditionally this takes days or weeks. Elicit automates the search and extracts key findings for you.

	Traditional Literature Review	Elicit AI Review
Search	Manual Google Scholar, keyword hunting	One research question → Elicit finds papers
Screening	Read abstract of every paper manually	AI ranks by relevance automatically
Extraction	Copy key facts from each paper manually	AI fills columns: methods, results, sample size
Synthesis	Write summary from notes	AI drafts synthesis across all papers



■ IMAGE PROMPT

Soft pastel: a research question in a search box at the top; below it a grid of paper cards ranked by relevance score; columns extracting methods, results, sample size from each paper; at the bottom a synthesis paragraph being assembled; cream background; minimal text; flat style.